

DHRUV ASHOK PATEL

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MS Cybersecurity Engineering student at USC with hands-on experience in security engineering, vulnerability assessment, and application security. Strong computer science fundamentals with a proven ability to build and ship security-focused software across network, mobile, and cloud environments. Published researcher in malware classification and cryptography with a passion for solving challenging, innovative security problems at scale.

EDUCATION

Master of Science in Cybersecurity Engineering | University of Southern California Aug 2025 – May 2027
GPA: 3.5/4.0 | Relevant Coursework: Foundations and Policy for Information Security; Computer Forensics; Security and Privacy; Malware Analysis and Reverse Engineering

B.E. Computer Science & Engineering (Cybersecurity) | University of Mumbai Dec 2021 – May 2025
GPA: 9.1/10 | Coursework: Fundamentals of Information Security, Digital Forensics, Cybersecurity Risk Management, Information Security Compliance, Ethical Hacking & Cybersecurity, ML for Cybersecurity, DL for Cybersecurity

SKILLS

- **Programming & Scripting:** Python, Java, C, C++, Bash, PowerShell, SQL
- **Security Engineering:** Vulnerability Assessment, Application Security, Threat Detection, Incident Response, Data Protection, Network Security, Mobile Security, Digital Forensics
- **Security Tools:** Wireshark, Burp Suite, Metasploit, Autopsy, EnCase, Splunk, Nessus
- **CS Fundamentals:** Algorithms, Data Structures, Operating Systems, Networking (TCP/IP, DNS, HTTP/HTTPS, Firewalls, Packet Analysis)
- **Cloud & Systems:** AWS (IAM, GuardDuty), Linux, Windows, Active Directory, Mobile App Development
- **Frameworks & Standards:** NIST, ISO 27001, MITRE ATT&CK, OWASP
- **Development & Automation:** Git, REST APIs, Security Automation, Log Analysis
- **Machine Learning (Applied):** Malware Classification, Feature Engineering, Model Evaluation

WORK EXPERIENCE

- Cybersecurity Intern | Crossover Technology Pvt Ltd.** Mar 2025 – Jun 2025
- Improved threat detection efficiency by 30% by developing Python-based security automation scripts to accelerate vulnerability triage and reduce mean-time-to-detect across security operations.
 - Performed manual vulnerability scans across internal systems, triaging findings by severity and communicating results to the security team to support prioritized remediation.
 - Participated in security team meetings, gaining exposure to real-world incident response workflows, threat discussion, and cross-functional security decision-making processes.

PROJECTS & RESEARCH

- **Security of LLM-Powered AI Agents – Prompt Injection & Data Exfiltration | Python, LLMs** 2025 – Present
Currently conducting independent security research on vulnerabilities in LLM-powered AI agents, focusing on prompt injection, multi-turn manipulation, and data exfiltration risks, with the goal of contributing toward a publication.
- **Malware Detection Using Deep Learning | Python, CNN, LSTM, TensorFlow** – Github
Achieved 98.74% malware detection accuracy by engineering a hybrid CNN-LSTM model, applying adversarial ML techniques to identify threats that evade traditional signature-based detection.
- **Quantum-Resistant Online Payment System | Python, FastAPI, MongoDB** – Github
Achieved 90%+ simulated reduction in key-interception risk by designing a QKD-inspired encrypted payment API, demonstrating expertise in data protection and secure application architecture.
- **Predictive Maintenance Using Machine Learning | Python, ML Algorithms** – Github
Achieved 98.5% accurate failure prediction by training supervised ML models on sensor data, demonstrating strong CS fundamentals and applied problem-solving skills.
- **TryHackMe Capture the Flag Challenges | Hands-on Security Labs**
Completed 50+ labs covering vulnerability assessment, intrusion detection, digital forensics, mobile security, and exploit fundamentals, reinforcing practical security engineering skills.

PUBLICATIONS

- **Advancing Security in Digital Transactions using Quantum Cryptography** – Journal of Information Systems Engineering and Management (JISEM), Apr 2025 – Read here
Proposed a QKD-inspired cryptographic framework for securing digital payment transactions, achieving a simulated 90%+ reduction in key-interception risk. Designed and implemented an encrypted API workflow using FastAPI and MongoDB demonstrating practical application of post-quantum security principles to real-world financial systems.
- **Malware Classification using Deep Learning** – International Research Journal of Engineering and Technology (IRJET), Oct 2024 – Read here
Evaluated six deep learning architectures (CNN, VGG16, VGG19, ResNet50, Xception, MobileNet) for malware family classification on the Maling dataset comprising 9,339 images across 25 malware families. MobileNet achieved the highest

accuracy at 98.74% with a loss of 0.0386, outperforming all other models. Used image-based malware visualization and texture analysis to classify malware binaries without requiring code execution or disassembly.

- **Predictive Maintenance for Industrial Equipment using Machine Learning** – International Journal of Advanced Research in Science, Communication and Technology (IJARSCT), Aug 2024 – [Read here](#)
Designed and evaluated supervised ML models trained on industrial sensor data to predict equipment failures with 98.5% accuracy, enabling proactive maintenance decisions and demonstrating applied skills in anomaly detection, data pipelines, and model evaluation.

CERTIFICATIONS

- Google Cybersecurity Specialization – Google
- Cybersecurity for Everyone – University of Maryland
- Introduction to Data Science in Python – University of Michigan

EXTRACURRICULAR ACTIVITIES

- 1st Place – WordHack Competition, CyberPeace Foundation
- Completed Social Internship conducted by the CyberPeace Foundation focused on cybersecurity awareness.
- Completed in-house internships in web development and mobile app development.